Lab 7
Getting Started with Operational Amplifier Circuits

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Date: October 15, 2026

I. Introduction

This lab had two main objectives. The first was to build and test a basic inverting amplifier using a 741 operational amplifier (op-amp), choosing resistor values for a gain of −10 and measuring that gain on the oscilloscope, including what happens when the input becomes large enough to push the output past the supply voltages. The second was to extend the same circuit into a summing inverting amplifier, using the DC power supply and a second input resistor to add a −1 V DC offset to the amplified sine wave, without needing a function generator that can produce an offset of its own.

II. Results

1. Building the Inverting Op-Amp Circuit

The inverting amplifier from the manual was built on the protoboard with the 741 op-amp straddling the center channel (Fig. 1). The op-amp was powered from the DC power supply with VCC+ = +10 V on pin 7 and VCC− = −10 V on pin 4, using the long bus strips of the protoboard for +10 V, −10 V, and ground. The non-inverting input (pin 3) and the ground of the function generator were both tied to the common node of the power supply, which served as the reference ground for the entire circuit.

To choose the resistors, the ideal op-amp assumptions were applied: no current flows into either input, and negative feedback forces Vn = Vp. Since pin 3 is grounded, Vn = Vp = 0 V. Writing KCL at the inverting input gives:

$$(V_n - V_{in})/R_i + (V_n - V_{out})/R_f = 0 \;\rightarrow\; -V_{in}/R_i - V_{out}/R_f = 0 \;\rightarrow\; V_{out}/V_{in} = -R_f/R_i$$

so a gain of −10 only requires Rf = 10Ri. Nominal values of Ri = 1 kΩ and Rf = 10 kΩ were selected. Measured with the DMM, these were Ri = 0.996 kΩ and Rf = 9.883 kΩ, giving a predicted gain of −9.883/0.996 = −9.92.

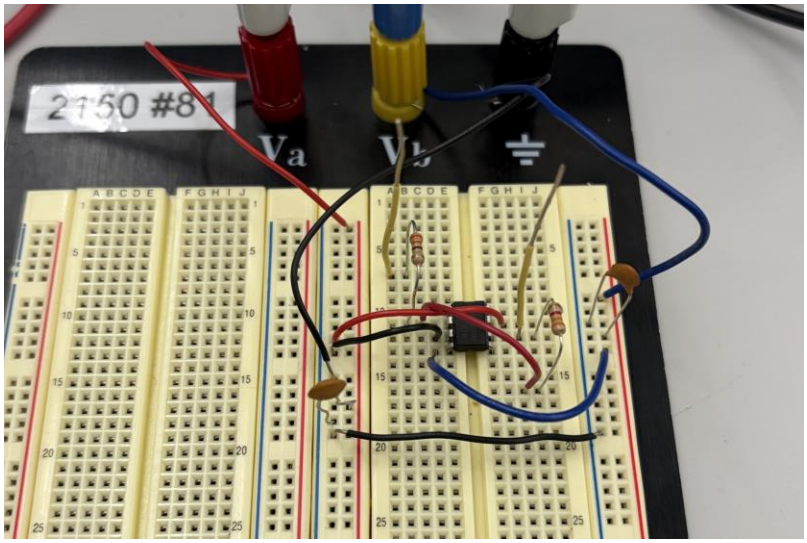


Figure 1: Inverting op-amp circuit built on the protoboard.

2. Measuring the Gain

The function generator was set to a 1000 Hz sine wave with a 0.5 V amplitude (1 Vpp) and no DC offset, with its output load set to High Z so that the output matched the set value. With the input displayed on channel 1 and the output on channel 2 of the oscilloscope, the input measured 1.01 Vpp and the output measured 9.49 Vpp. The measured gain was therefore:

$$G = Vout/Vin = -9.49/1.01 = -9.40$$

This agrees reasonably well with the predicted value of −9.92 (and the design value of −10), differing by about 5%. The difference comes from the measured resistors not being exactly 1 kΩ and 10 kΩ, the 741 not being an ideal op-amp, and the limited precision of reading amplitudes off the oscilloscope.

The negative sign of the gain was verified by comparing the two signals in time on the oscilloscope. The output was 180° out of phase with the input: whenever the input reached its positive peak, the output reached its negative peak, and vice versa (the same inversion can be seen in Fig. 3). A positive gain would instead have produced an output that peaked at the same time as the input.

3. Increasing the Input Amplitude

Once the circuit was confirmed to be working, the input amplitude was increased to 1.2 V (2.4 Vpp). The input measured 2.41 Vpp and the output measured 20.9 Vpp, a gain of only −8.67 compared with −9.40 in section 2. At a gain of −9.40, a 2.41 Vpp input should have produced about 22.7 Vpp, or roughly ±11.3 V, which is beyond the ±10 V supply rails. An op-amp's output cannot swing past its supply voltages, so once the input became large enough, the output could no longer follow it, and the gain dropped. This shows that the gain formula −Rf/Ri only holds while the output stays within the linear range of the op-amp, which is set by its supply voltages.

4. Designing the Summing Inverting Amplifier

To add a −1 V DC offset to the output without a function generator offset, the circuit was extended into a summing inverting amplifier by adding a second input resistor, R2, from a DC source v2 to the inverting input (the summing point). With the same ideal op-amp assumptions, the output is the sum of each input scaled by its own gain:

$$vo = -(Rf/R1)v1 - (Rf/R2)v2$$

where v1 and R1 are the original sine wave input and input resistor from section 1. Since the DC power supply was already powering the op-amp, v2 had to be either +10 V or −10 V. Because the DC term is inverted by the amplifier, the +10 V supply was used, and R2 was chosen so that the DC term equals −1 V:

$$-(Rf/R2)(+10\ V) = -1\ V \ \text{--> } R2 = 10Rf = 100\ k\Omega$$

In other words, the DC input sees a gain of −Rf/R2 = −0.1, which scales the +10 V supply down to −1 V at the output, while the sine wave still sees the original gain of −Rf/R1 = −10. With these values, the expected output was:

$$vo = -(10/1)v1 - (10/100)v2 = -10v1 - 1\ V$$

The modified circuit is shown in Fig. 2.

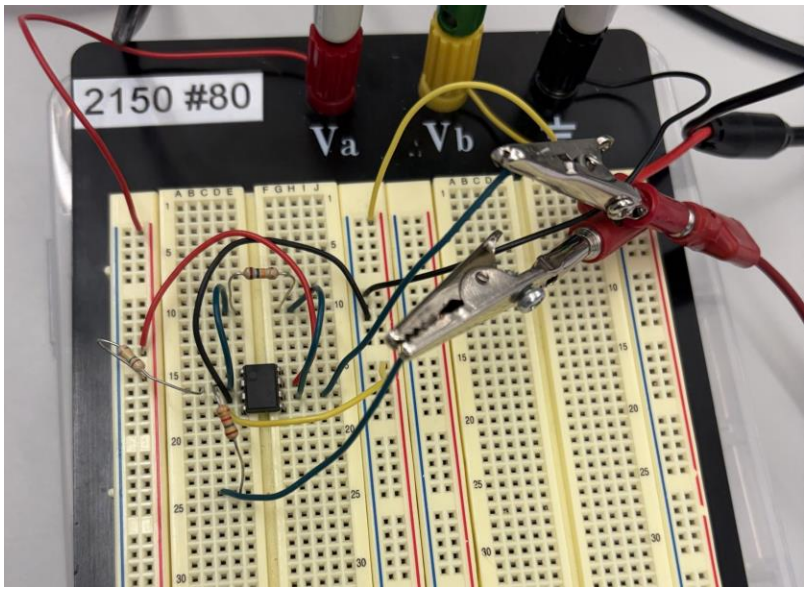


Figure 2: Summing inverting amplifier built on the protoboard.

5. Adding the −1 V DC Offset

The function generator was again set to a 1000 Hz, 0.5 V amplitude sine wave with no DC offset, and both signals were displayed on the oscilloscope (Fig. 3). The input on channel 1 was unchanged from section 2. The output on channel 2 was still an inverted 1 kHz sine wave, but it was no longer centered at 0 V. Using the oscilloscope cursors, the output peak was at 4.06 V and the trough was at −6.31 V, a swing of 10.37 Vpp centered at (4.06 + (−6.31))/2 ≈ −1.13 V.

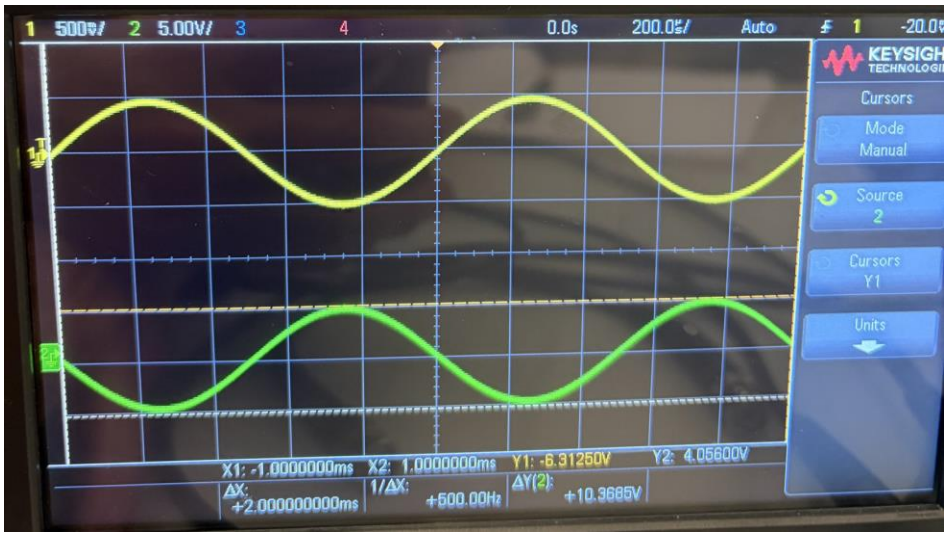


Figure 3: Summing amplifier input and output at 200 µs/div, with cursors on the output peak and trough.

This is what was expected. By superposition, the AC part of the output should be the same inverted, roughly 10× sine wave from section 2, and the DC part should shift the entire waveform down by 1 V, so an output swinging about ±5 V around −1 V (from about +4 V to −6 V) was predicted. The measured center of −1.13 V is close to the −1 V target. The small

differences in both the offset and the 10.37 Vpp swing, compared with 9.49 Vpp in section 2, are mostly due to the cursors being placed by hand at 5 V/div, where each minor division is 1 V and the trace thickness alone makes the exact peak hard to pin down, along with the tolerance of R2 and the supply not being exactly 10.00 V. Since the output stayed within the supply rails, the AC and DC parts added together without any distortion.

III. Discussion and Conclusions

This lab showed how an op-amp with negative feedback can be used to build a predictable amplifier from nothing more than two resistors. Using the ideal op-amp assumptions (Vn = Vp and no current into the inputs), the gain of the inverting amplifier reduced to −Rf/Ri, and choosing Rf = 10Ri produced a measured gain of −9.40, within about 5% of the value predicted from the measured resistors. The negative sign appeared directly on the oscilloscope as a 180° phase inversion between the input and output. Increasing the input to 1.2 V showed the main practical limit of the circuit: the op-amp cannot produce an output beyond its supply voltages, so the gain only holds while the output stays within that range.

The summing amplifier extended the same idea. Since each input to the summing point is scaled by its own gain, a DC supply that was already on the bench could be turned into a −1 V offset just by choosing R2 = 10Rf, and the measured output, a 10.37 Vpp sine wave centered near −1.13 V, matched the superposition prediction. Overall, this lab connected the ideal op-amp analysis from class to real measurements, thus demonstrating two ideas: setting gain with a resistor ratio and summing signals by superposition, both bounded by the supply rails of the op-amp.

IV. References

[1] Dr. Iman Salama. “Lab 7 – Getting Started with Operational Amplifier Circuits.” Northeastern University. 10 October 2023.